Time and Temperature Table for **Cooked Beef** (not Poultry)

Cooked beef and roast beef, including sectioned and formed roasts, chunked and formed roasts, and cooked corned beef can be prepared using one of the following time and temperature combinations to meet either a 6.5-$\log_{10}$ or 7-$\log_{10}$ reduction of *Salmonella*. The stated temperature is the minimum that must be achieved and maintained in all parts of each piece of meat for a least the stated time.

These guidelines, if followed precisely, are a validated process for HACCP plans.

Minimum Internal Temperature		Minimum processing time in minutes or seconds after minimum temperature is reached	
Degrees Fahrenheit	**Degrees Centigrade**	**6.5-$\log_{10}$ Lethality**	**7-$\log_{10}$ Lethality**
130	54.4	112 min.	121 min.
131	55.0	89 min.	97 min.
132	55.6	71 min.	77 min.
133	56.1	56 min.	62 min.
134	56.7	45 min.	47 min.
135	57.2	36 min.	37 min.
136	57.8	28 min.	32 min.
137	58.4	23 min.	24 min.
138	58.9	18 min.	19 min.
139	59.5	15 min.	15 min.
140	60.0	12 min.	12 min.
141	60.6	9 min.	10 min.
142	61.1	8 min.	8 min.
143	61.7	6 min.	6 min.
144	62.2	5 min.	5 min.
145	62.8	4 min.	4 min.
146	63.3	169 sec.	182 sec.
147	63.9	134 sec.	144 sec.
148	64.4	107 sec.	115 sec.
149	65.0	85 sec.	91 sec.
150	65.6	67 sec.	72 sec.
151	66.1	54 sec.	58 sec.
152	66.7	43 sec.	46 sec.
153	67.2	34 sec.	37 sec.
154	67.8	27 sec.	29 sec.
155	68.3	22 sec.	23 sec.
156	68.9	17 sec.	19 sec.
157	69.4	14 sec.	15 sec.
158	70.0	0 sec.	0 sec.
159	70.6	0 sec.	0 sec.
160	71.1	0 sec.	0 sec.

Adapted from USDA FSIS Appendix A: Compliance Guidelines For Meeting Lethality Performance Standards For Certain Meat And Poultry Products. www.fsis.usda.gov/OPPDE/rdad/FRPubs/95-033F/95-033F_Appendix%20A.htm

Time and Temperature Tables for **Cooking Poultry**

The holding times for cooking poultry for a 7-$\log_{10}$ reduction of *Salmonella* are longer than the cooking time for other meats. Poultry with different amounts of fat require different cooking times.

Times for given temperature, fat level, and species needed to obtain 7-$\log_{10}$ lethality of *Salmonella*		
Fat%=2		
Temperature (°F)	**Time for Chicken**	**Time for Turkey**
136	64.5 min	64.3 min
137	51 min	52.5 min
138	40.5 min	42.5 min
139	32.2 min	34.6 min
140	25.7 min	28.3 min
141	20.5 min	23.2 min
142	16.4 min	19 min
143	13.2 min	15.6 min
144	10.6 min	12.8 min
145	8.6 min	10.6 min
146	6.9 min	8.7 min
147	5.5 min	7.1 min
148	4.4 min	5.8 min
149	3.5 min	4.7 min
150	2.7 min	3.7 min
151	2 min	2.9 min
152	1.5 min	2.3 min
153	1.2 min	1.8 min
154	56.9 sec	1.5 min
155	45 sec	1.2 min
156	35.6 sec	59.3 sec
157	28.2 sec	48.1 sec
158	22.3 sec	39 sec
159	17.6 sec	31.7 sec
160	14 sec	25.7 sec
161	11 sec	20.9 sec
162	<10.0 sec	16.9 sec
163	<10.0 sec	13.7 sec
164	<10.0 sec	11.2 sec
165	<10.0 sec	<10.0 sec

Times for given temperature, fat level, and species needed to obtain 7-$\log_{10}$ lethality of *Salmonella*		
Fat%=4		
Temperature (°F)	**Time for Chicken**	**Time for Turkey**
136	67 min	64.9 min
137	53.2 min	52.8 min
138	42.2 min	43 min
139	33.6 min	35.1 min
140	26.8 min	28.7 min
141	21.5 min	23.5 min
142	17.2 min	19.3 min
143	13.8 min	15.9 min
144	11.1 min	13 min
145	8.9 min	10.7 min
146	7.2 min	8.8 min
147	5.7 min	7.2 min
148	4.5 min	5.8 min
149	3.6 min	4.7 min
150	2.7 min	3.7 min
151	2.1 min	2.9 min
152	1.6 min	2.3 min
153	1.2 min	1.9 min
154	59.1 sec	1.5 min
155	46.8 sec	1.2 min
156	37 sec	59.8 sec
157	29.3 sec	48.5 sec
158	23.2 sec	39.4 sec
159	18.3 sec	32 sec
160	14.5 sec	26 sec
161	11.5 sec	21.1 sec
162	<10.0 sec	17.1 sec
163	<10.0 sec	13.9 sec
164	<10.0 sec	11.3 sec
165	<10.0 sec	<10.0 sec

Times for given temperature, fat level, and species needed to obtain 7-$\log_{10}$ lethality of *Salmonella*		
Fat%=6		
Temperature (°F)	**Time for Chicken**	**Time for Turkey**
136	69.9 min	65.8 min
137	55.5 min	53.6 min
138	44.2 min	43.8 min
139	35.2 min	35.8 min
140	28.2 min	29.3 min
141	22.6 min	24.1 min
142	18.1 min	19.8 min
143	14.6 min	16.3 min
144	11.8 min	13.4 min
145	9.5 min	11 min
146	7.6 min	9 min
147	6.1 min	7.4 min
148	4.8 min	6 min
149	3.8 min	4.8 min
150	2.9 min	3.8 min
151	2.1 min	2.9 min
152	1.6 min	2.3 min
153	1.3 min	1.9 min
154	1 min	1.5 min
155	48.6 sec	1.2 min
156	38.4 sec	1 min
157	30.4 sec	49 sec
158	24 sec	39.8 sec
159	19 sec	32.3 sec
160	15 sec	26.2 sec
161	11.9 sec	21.3 sec
162	<10.0 sec	17.3 sec
163	<10.0 sec	14 sec
164	<10.0 sec	11.4 sec
165	<10.0 sec	<10.0 sec

Times for given temperature, fat level, and species needed to obtain 7-$\log_{10}$ lethality of *Salmonella*		
Fat%=8		
Temperature (°F)	**Time for Chicken**	**Time for Turkey**
136	73 min	66.9 min
137	58.2 min	54.7 min
138	46.4 min	44.7 min
139	37.2 min	36..7 min
140	29.8 min	30.2 min
141	24 min	24.9 min
142	19.4 min	20.5 min
143	15.6 min	17 min
144	12.6 min	14 min
145	10.2 min	11.5 min
146	8.2 min	9.5 min
147	6.6 min	7.7 min
148	5.2 min	6.3 min
149	4.1 min	5 min
150	3.1 min	4 min
151	2.3 min	3.1 min
152	1.7 min	2.3 min
153	1.3 min	1.9 min
154	1.1 min	1.5 min
155	45 sec	1.3 min
156	50.4 sec	1 min
157	39.9 sec	49.5 sec
158	31.6 sec	40.1 sec
159	19.8 sec	32.6 sec
160	15.6 sec	26.4 sec
161	12.4 sec	21.5 sec
162	9.4 sec	17.4 sec
163	<10.0 sec	14.1 sec
164	<10.0 sec	11.5 sec
165	<10.0 sec	<10.0 sec

Times for given temperature, fat level, and species needed to obtain 7-$\log_{10}$ lethality of *Salmonella*		
Fat%=10		
Temperature (°F)	**Time for Chicken**	**Time for Turkey**
136	76.7 min	68.4 min
137	61.4 min	56.2 min
138	49.2 min	46.2 min
139	39.6 min	38.1 min
140	32 min	31.5 min
141	25.9 min	26.2 min
142	21 min	21.7 min
143	17.1 min	18 min
144	13.9 min	15 min
145	11.3 min	12.4 min
146	9.1 min	10.2 min
147	7.4 min	8.4 min
148	5.8 min	6.8 min
149	4.6 min	5.4 min
150	3.5 min	4.3 min
151	2.6 min	3.3 min
152	1.9 min	2.5 min
153	1.4 min	1.9 min
154	1.1 min	1.6 min
155	52.4 sec	1.3 min
156	41.4 sec	1 min
157	32.8 sec	49.9 sec
158	25.9 sec	40.5 sec
159	20.5 sec	32.9 sec
160	16.2 sec	26.7 sec
161	12.8 sec	21.7 sec
162	10.2 sec	17.6 sec
163	<10.0 sec	14.3 sec
164	<10.0 sec	11.6 sec
165	<10.0 sec	<10.0 sec

Adapted from USDA FSIS, "Time-Temperature Tables for Cooking Ready-To-Eat Poultry Products"
www.fsis.usda.gov/wps/wcm/connect/9ab2e062-7ac8-49b7-aea1-f070048a113a/RTE_Poultry_Tables.pdf?MOD=AJPERES